BA - TASK - 5

Project Name:
Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Introduction:
This project develops a financial risk simulator for a portfolio containing multiple assets. It uses historical asset returns, covariance analysis, Cholesky decomposition and Monte Carlo simulation to estimate the 99% Value-at-Risk (VaR) of the portfolio.

Objectives:

- Analyze risk across 500 assets
- Calculate the historical covariance matrix
- Generate correlated random returns
- Simulate future asset prices using GBM
- Calculate portfolio 99% VaR

Data Preparation & Covariance Matrix

Step 1: Historical Data

Historical price data for 500 assets is collected and converted into daily returns.

import pandas as pd

import numpy as np

```
data = pd.read_csv("asset_prices.csv")


returns = data.pct_change().dropna()


cov_matrix = returns.cov()


print(cov_matrix.shape)
```

Output:

(500, 500)

Explanation:
The covariance matrix shows how the returns of different assets move in relation to each other.

Cholesky Decomposition & GBM

Step 2: Cholesky Decomposition

```
L = np.linalg.cholesky(cov_matrix)


random_noise = np.random.normal(
    0, 1, (500, 1000)
)


correlated_noise = L @ random_noise
```

Step 3: Geometric Brownian Motion

```
S = S0 * np.exp(
    (mu - 0.5 * sigma**2) * dt
    + sigma * correlated_noise * np.sqrt(dt)
)
```

Explanation:
Cholesky decomposition introduces the relationships between assets, while GBM is used to simulate possible future asset-price paths.

Results & Conclusion :

Results:

- 500 assets are analyzed.
- A **500 × 500 covariance matrix** is generated.
- Cholesky decomposition creates correlated random variables.
- Multiple future portfolio scenarios are simulated using Monte Carlo.
- The portfolio's **99% VaR** is calculated.

Visualization:

```
import matplotlib.pyplot as plt


plt.hist(portfolio_returns, bins=50)
```

```
plt.axvline(-VaR_99)

plt.xlabel("Portfolio Returns")

plt.ylabel("Frequency")

plt.title("Monte Carlo Portfolio Risk")

plt.show()
```

Conclusion:

The project provides a practical way to measure portfolio risk using historical data and Monte Carlo simulation. The 99% VaR gives an estimate of the potential portfolio loss under simulated market conditions and helps in better risk management.